International Journal of Science, Mathematics and Technology Learning
ISSN: 2327-7971 (Print) ISSN: 2327-915X (Online)
Volume 32 No. 1, 2024

Natural Products: Classification and Role in the Discovery of Important Drugs

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Abstract

Natural products and their structural analogues have enormous contribution to drug discovery and development. The success stories for the discoveries of Penicillin from Penicillium rubens, Aspirin from Willow bark of Salix alba tree and Taxol from yew tree etc. have pave the way for researchers in the field of natural product chemistry to focus their research work in drug discovery and development from plants and microorganisms. Drug discovery involves the identification of new chemical entities of potential therapeutic value, which can be obtained from natural sources through isolation and by chemical synthesis or a combination of both. Although there are some challenges for drug discovery, such as technical barriers to screening, isolation, characterization and optimization but several technological and scientific developments including improved isolation, analytical tools, genome engineering strategies, and microbial culturing advances are addressing such challenges and opening up new opportunities for drug discovery process. In this review role of natural products in the discovery important drugs are highlighted.

Keywords.

Natural Products, Metabolites, Drug Discovery, Pharmaceuticals, Biological Activity.

Introduction

Natural products are chemical compounds produced by living organisms including plants, invertebrates and microorganisms [1]. They are also known as secondary metabolites which

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Acceptance: 16April2024, Publication: 20April2024

are not essential for life but they play key roles in biological activities. Although traditional medicine has a long history of clinical practice in many countries due to the fact that nature provides a rich source of active ingredients for multi-target therapies. However, insufficient study of clinical efficacy, safety, potency and their purity have been great concerns [2]. Some components in natural product mixtures may cause severe adverse effects and chronic toxicities with carcinogenic and endocrine-disrupting effects [3]. To avoid unpredictable adverse side effects and to get potential drug candidates, scientists all over the world have committed intensive efforts to isolating and identifying bioactive pure compounds from nature with the help of numerous modern techniques. Lead compounds have pharmacological or biological activities likely to be therapeutically useful and they are leading candidates for new drugs. The studies of the mechanisms of action and toxicity of pure leads isolated from natural resources at molecular levels are very important, which will give the direction for lead improvement and further drug development.

Despite the changes in drug discovery strategies and most notably the emergence of medicines derived from molecular biology, there remains a need to develop natural product-based medicines which has shown great success as a strategy in the diversified drug discoveries for the human use [4]. Natural products have been used over the years throughout human history and evolution as a rich and valuable source of drugs, bioactive compounds and modern pharmaceuticals for various remedies and ailments [5, 6]. Most of the drugs have been discovered using natural products and related compounds [7, 8]. Some natural product-derived drugs are a hallmark of modern pharmaceuticals which include quinine, penicillin G, morphine, paclitaxel, digoxin and many other examples including antibiotics [9, 10]. Natural products offer special advantages and challenges for the drug discovery process in comparison with conventional synthetic molecules. Natural products are characterized by enormous scaffold diversity and structural complexity. Starting from the late 19th century, scientists and physicians had been very much interested about the antibacterial properties of the different types of moulds including the mould penicillium but they were unable to find out what process was causing the effect. The effect of mould penicillium with antibacterial activities was finally discovered in 1928 by Scottish scientist Alexander Fleming [11]. Similarly, recent literature also revealed how drugs were reportedly discovered from natural products. There were very few suitable medical products available for the treatment of any diseases. In fact, many of the drugs available today contain active ingredients extracted from natural products. Natural product drug discovery continues to play a significant role in the clinical development of new therapies in the biopharmaceutical industry. Most of the small molecule anticancer drugs are natural product-based or natural product inspired [12]. Drug discovery involves the identification of new chemical entities of potential therapeutic value, which can be obtained through isolation from natural sources and via chemical synthesis or a combination of both. In the conventional approach drug targets were exposed to crude extracts and in case of evidence of pharmacological activity the extract was fractionated and the active compound isolated and identified. This method was slow, laborious, inefficient, and provided no guarantee that a lead from the screening process would be chemically workable or even patentable. As natural products usually are molecules with more complex structures, it was more difficult to extract, purify, isolate or synthesize in sufficient quantities for discovery and evaluation of pharmacological activities [13]. Drug discovery is leading to be a challenging scientific task to find robust and viable lead candidates, which is nothing but the process flow from a screening of isolated natural products or synthetic compounds that requires expertise and experience [14]. However, in addition to their chemical structure diversity and their biodiversity, the development of new technologies has revolutionized the screening of natural products in discovering new drugs [15]. These technologies offer a unique opportunity to re-establish natural products as a major source for drug discovery.

Natural Products Classification

Plants, animals and microorganisms have been the most important sources for biologically active natural products. Terrestrial and aquatic species of plants and microorganisms, especially those of marine origin, produce unique bioactive substances of valuable therapeutics and lead structures for potential new drugs. More than 200,000 structures of natural products are known only some selected groups and important compounds are discussed in this article. Natural products are classified into different groups according to their biosynthetic origin. The major classes are alkaloids, terpenoids, steroids, flavonoids and other miscellaneous compounds [16].

Alkaloids

Alkaloids are nitrogenous heterocyclic organic compounds mostly isolated from plants and biosynthetically are derived from amino acids resulting in variety of chemical compounds [17, 18]. Alkaloids account for approximately 20% of the reported secondary metabolites from plants [19]. More than 3,000 different types of alkaloids have been identified from plant species [20]. In plants, alkaloids play an essential role in their natural defense and growth [21]. Alkaloids extraction and processing have been major areas of research and development [22, 23]. Alkaloids play an essential role in both human medicine and in an organism's natural defence. Well known alkaloids (Figure1 & Figure 3) include morphine, strychnine, quinine, ephedrine, vincristine, vinblastine, nicotine, Ibogaine and many more with array of important biological activities [24].

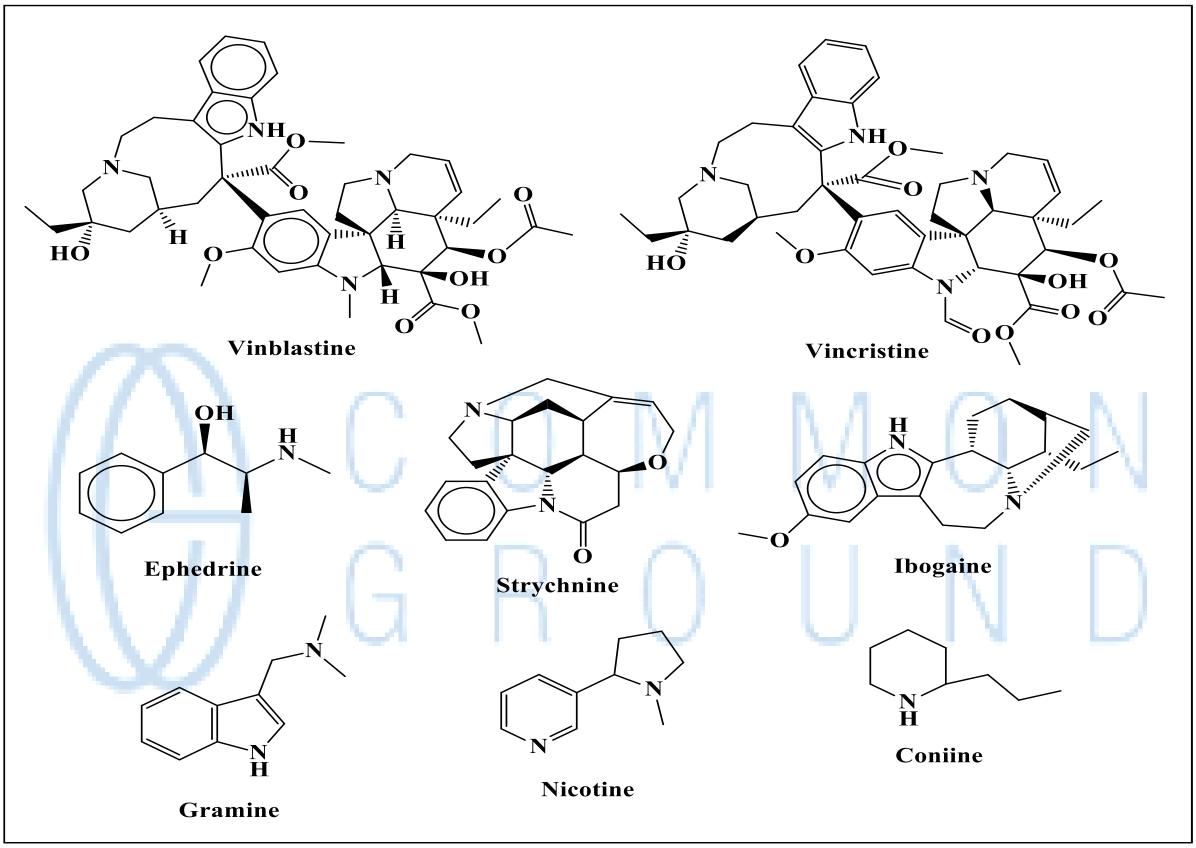

Figure1: Example of some alkaloids

Terpenoids

Terpenes and terpenoids are widely found in nature and are the largest group of natural compounds found in the plants. Terpenes are simple hydrocarbons structures which can be classified as monoterpenes, sequiterpenes, diterpenes, sesterpenes, triterpenes, tetraterpenes and poly terpenes based on the number of building blocks present. The larger structures are assembled from several isoprene units by linking head-to-tail isoprene units (e.g. ß-Carotene). Terpenes can be cyclic or acyclic, with a large range of structural diversity [25]. Some examples of terpenes are (Figure 2): pinene, myrcene, limonene, terpinene, *p*-cymene, citronelle, humulene [26].

Terpenoids are modified class of terpenes containing different functional groups such as alcohols, aldehydes, esters, ether, epoxides, ketones, phenols and oxidized methyl groups at various positions. Some examples of terpenoids are: citronellal, geraniol, carvacrol, linalool, linalyl acetate, thymol, and menthol [27]. These compounds have several biological activities such as anticancer, anti-allergic, antibacterial and antioxidant activities [28-30]. Some botanical compounds, such as carvone, menthol, ascaridol, methyl eugenol, toosendanin, azadirachtin, and volkensin, have been shown to exhibit antimicrobial and antifungal properties, as well as insect pest repellent properties [31]. The biosynthetic pathway for the synthesis of terpenes and terpenoids involve the mevalonic acid (MVA) pathway in the cytosol and the 2C-methyl-D-erythritol-4- phosphate (MEP) pathway in the plastid for the formation of precursors, isopentenyl pyrophosphate (IPP) and dimethylallyl pyrophosphate (DMAPP) [32]. Examples of bioactive compounds of terpenes and terpenoids are presented (Figure 2).

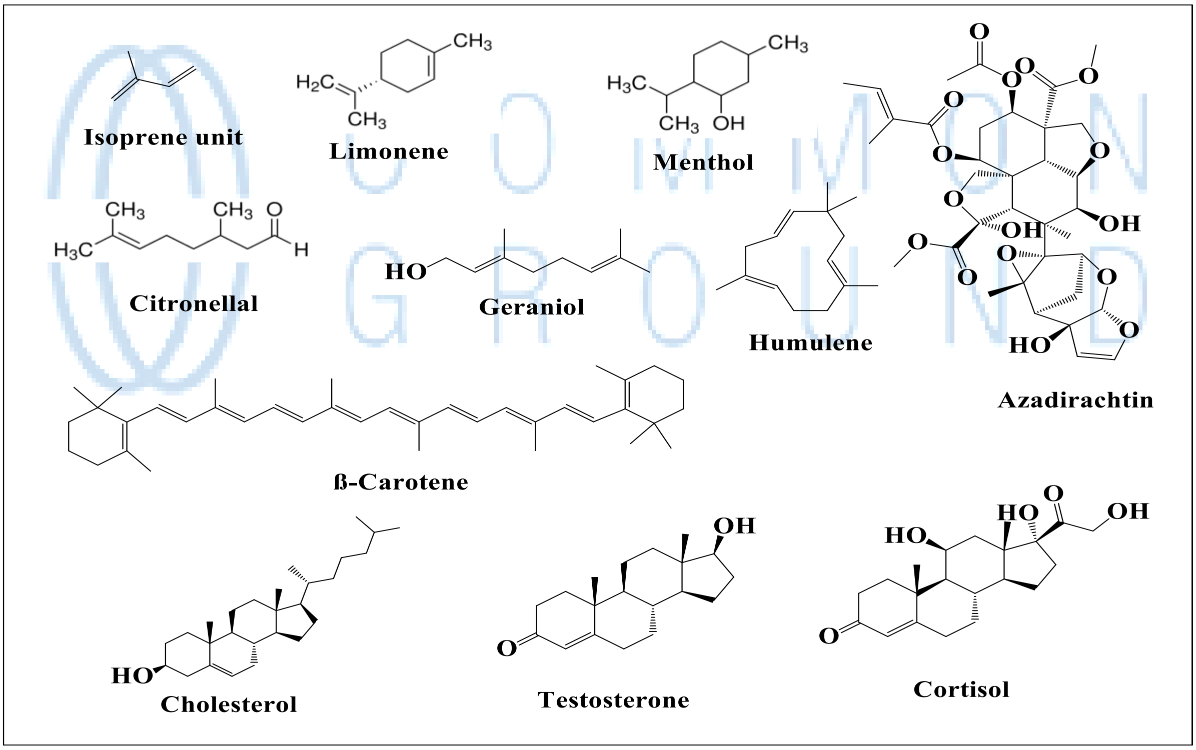

Figure 2: Examples of terpenoids and steroids

Steroid

Steroid is another class of organic compound that contains a characteristic arrangement of four cycloalkane rings joined to one another. The steroid core is composed of seventeen carbon atoms bonded together in the form of four fused rings: three cyclohexane rings designated and one cyclopentane ring [33]. The biosynthesis of steroid involves the mevalonate pathway which uses acetyl-CoA as building blocks to form dimethylallyl pyrophosphate (DMAPP) and isopentenyl pyrophosphate (IPP) in humans and other animals. These two compounds are combined to form geranyl pyrophosphate (GPP) which is then converted to lanosterol, the first steroid in the pathway via squalene after oxidation and multiple rearrangements. From lanosterol further reactions produces different steroids [34].

The primary use of natural steroids in health care is to reduce inflammation and other disease symptoms. Steroids make the whole immune system less active, which can be very useful in illnesses treatment. Examples of steroids include the dietary lipid cholesterol, the sex hormones estradiol and testosterone, bile acids, and drugs such as the anti-inflammatory agent dexamethasone etc. (Figure 2) [35]. Cholesterol, which modulates the fluidity of cell membranes and is the principal constituent of the plaques implicated in atherosclerosis. Cortisol commonly used is as the antiinflammatory medicine. Cortisol is used in ointments to treat skin ailments and in inhalers to treat asthmatic attacks. Progesterone controls events during pregnancy, and estradiol regulates female characteristics. Both of these steroid hormones are components of birth control pills and function by preventing ovulation [36, 37]

Flavonoids

Flavonoids are most commonly found in fruits, vegetables, flowers, herbs, cereals, nuts, and seeds which mainly consist of a benzopyrone ring with phenolic or poly-phenolic groups at different positions [38-40]. Many flavonoid compounds have been isolated and identified [41]. Aurone, Luteolin, Kaempferol, and Myricetin are some well known flavonoids which exhibits important biological activities such as anticancer and anti-inflammatory activities. [42-45]. Most of the flavonoids are widely accepted as therapeutic agents [46]. Flavonoids are naturally synthesized via the phenylpropanoid pathway with bioactivity dependent on its absorption mechanism and bioavailability [47].

Role in drug discovery

Natural products have pharmacological activity that can be of therapeutic benefit in treating diseases. Natural products are the active components of many traditional medicines. Furthermore, synthetic analogues of natural products with improved potency and safety can be prepared and therefore natural products are often used as starting points for drug discovery. Natural product constituents have inspired numerous drug discovery efforts that eventually gained approval as new drugs by the U.S. Food and Drug Administration.

Indigenous peoples and ancient civilizations experimented with various plant and animal parts to determine what effect they might have. Through trial and error in isolated cases, traditional healers or shamans found some sources to provide therapeutic effect, representing knowledge of a crude drug that was passed down through generations in such practices as traditional medicine and ayurveda. Extracts of some natural products led to modern discovery of their active ingredients and eventually to the development of new drugs.

Drug is any chemical substance, natural or man-made (usually excluding nutrients, water, or oxygen), that by its chemical nature alters biological structure or functioning when administered and absorbed [48]. Drug can also be defined as a substance that could bring about a change in the biological functions through its chemical actions (Okeye, 2001). Comprehensively, drugs are also considered as a chemical that modifies the living tissues that could bring about psychological and behavioural changes (Olalekan *et al*, 2014). Moreover, a drug is defined as a substance that modifies perceptions, cognition, mood, behaviour and general body function (Balogun, 2006).

Drugs Classification

Drugs can be categorized in a number of ways. In the world of medicine and pharmacology, a drug can be classified as Narcotic analgesics, Depressants, Stimulants, Hallucinogens and Inhalants by its chemical activity or by the condition that it treats. Narcotic analgesics are a class of drugs that are used to get relieve from pain, induce euphoria, and create mood changes in the user. They may also be called opiates, opioid analgesics, or narcotics. Examples of narcotic analgesics include codeine, heroin, demerol, morphine, methadone and oxycontin.

Depressants type of drugs suppresses or slows the activity of the brain and nerves, acting directly on the central nervous system to create a calming or sedating effect. Depressants are taken to relieve anxiety, promote sleep and manage seizure activity. This category includes barbiturates (phenobarbital, thiopental, butalbital), benzodiazepines (alprazolam, diazepam, clonazepam, lorazepam, midazolam), alcohol, and gamma hydroxybutyrate.

On the other hand Stimulants works by accelerating the activity of the central nervous system. Stimulants can make you feel energetic, focused, and alert. This class of drugs can also make you feel edgy and angry. Stimulants include drugs such as cocaine, amphetamine, and methamphetamine. Amphetamine-derived stimulants like ecstasy and methamphetamine are the most commonly abused drugs around the world (Strobel, 2003).

Hallucinogen drugs act on the central nervous system to alter your perception of reality, time, and space. Hallucinogenic drugs include psilocybin, lysergic acid diethylamide and dimethyltryptamine. Hallucinogens may cause you to imagine situations that are not real.

Inhalants are class of drugs being primarily consumed through inhalation. These substances can be found in household items such as glue, paint thinners, gasoline, marker or pen ink, and others. Though ultimately all of these substances cross through the lungs into the bloodstream, their precise method of abuse may vary but can include sniffing, spraying, huffing, bagging, and inhaling, among other delivery routes.

Drugs derived from plant based natural products

A large number of currently prescribed drugs have been either directly derived from or inspired by plants natural products. A few representative examples are listed below. Some of the oldest natural product based drugs are analgesics. The bark of the willow tree has been known from antiquity to have pain relieving properties. This is due to the presence of the natural product salicin which in turn may be hydrolyzed into salicylic acid. A synthetic derivative acetylsalicylic acid better known as aspirin is a widely used for the treatment of pain, inflammation, and fever. Its mechanism of action is inhibition of the cyclooxygenase (COX) enzyme. Another notable example is opium which is extracted from the latex from Papaver somniferum (a flowering poppy plant). The most potent narcotic component of opium is the alkaloid morphine which acts as an opioid receptor agonist (Figure 3).

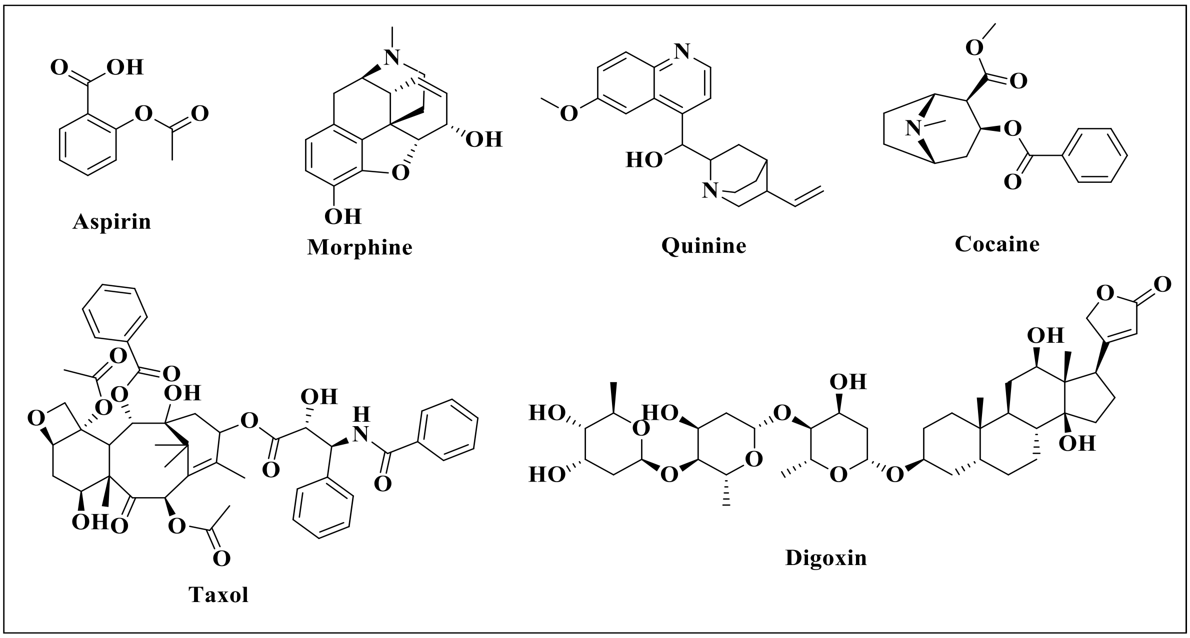

Figure 3: Drugs derived from plant sources

Quinine is also an alkaloid isolated from the Cinchona calisaya tree from South America and is used as one of the primary treatments for malaria. Although the drug can be synthesized, the most economically viable method of production is to extract it from the cinchona tree. Unfortunately, there are many side effects to this medication, including permanent kidney damage. Cocaine is another alkaloid obtained from the leaves of Erythroxylon coca, which belong to the family Erythroxylaceae. It is used as a local anesthetic as well for the digestive and nervous system. Paclitaxel (Taxol) is a chemotherapeutic agent derived from the bark of the Pacific Yew tree (Taxus brevifolia). Paclitaxel works by inhibiting a compound called tubulin inside the cells. Tubulin is needed during cell division, by inhibiting this compound, Paclitaxel is able to prevent cell division in the body. Since cancer often involves rapidly dividing cells, the drug is used to target these rapidly dividing cells more specifically, slowing the growth of cancer. Unfortunately, there are also a wide range of side effects, mostly as a result of the inhibition of rapidly dividing cells. Digoxin is a heart medication used for heart failure, and cardiac arrhythmias (such as atrial flutter, or atrial fibrillation). It was isolated from the foxglove plant (Digitalis lanata) in 1930, however, the first mention of foxglove derivatives for cardiac related conditions goes all the way back to 1785. It was used to treat a condition known as dropsy, which is an accumulation of fluid under the skin, often as a result of chronic heart failure. Digoxin works by inhibiting sodium, potassium, and ATP channels in the heart. This causes sodium levels to build up inside the cells of the heart, which then causes calcium to increase inside the cell as well. This heightened calcium allows stronger contraction of the heart, while expending less energy. It also has an effect on the vagus nerve, which is used to control the heart rhythm.

Drugs derived from microorganism and animal based natural products

Microorganisms were identified as sources of valuable natural products. Historically, microorganisms (mostly bacteria and fungi) have played an important role in providing new structures, like antibiotics for drug discovery and development. The terrestrial microbial populations are immensely diverse which is also reflected in the number of compounds and metabolites isolated from these microorganisms. As mentioned above, the similarity of many compounds from marine invertebrates like sponges, ascidians, soft corals and bryozoans to those isolated from terrestrial microbes led to the hypothesis that associated microorganisms might be responsible for their production. Over time it became more and more evident that a significant number of marine natural products are actually not produced by the originally assumed invertebrate but rather by microbes living in the symbioses with their invertebrate host [49].

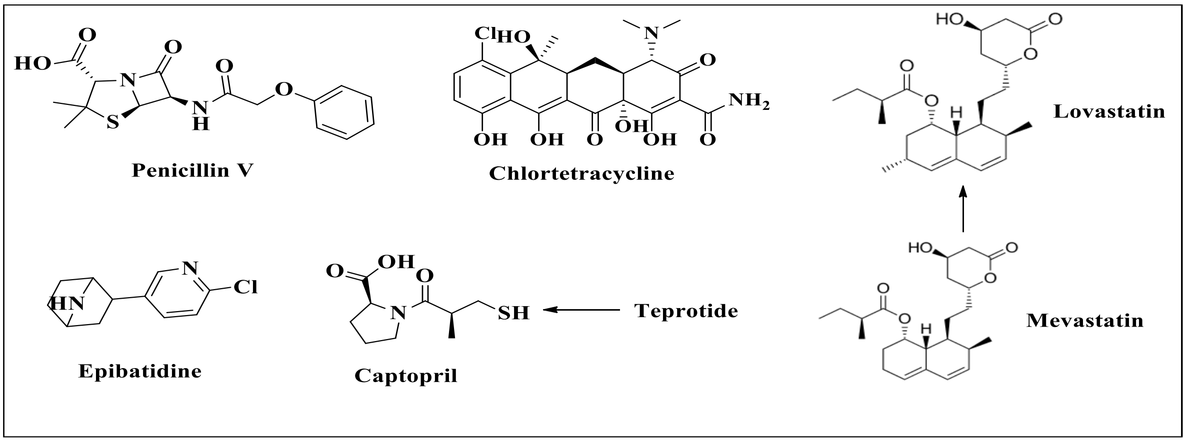

Figure 4: Some drugs derived from microorganisms and animals

The first antibiotic penicillin was discovered from the fungus *Penicillium rubens* by Alexander Fleming in 1928. Although there are many penicillins, they are all classified as beta-lactam antibiotics. They work by blocking a key process during cell division (peptidoglycan cross-link formation), causing the cell to eject its insides, effectively killing the bacteria. Tetracyclines are another class of antibiotics used to treat infections. Chlortetracycline is a member of the class of tetracyclines isolated from Streptomyces aureofaciens. It has a role as an antiprotozoal drug, a fluorescent probe, a calcium ionophore and an antibacterial drug. Another class of drugs widely used to lower cholesterol are the HMG-CoA reductase inhibitors, for example Lovastatin. These were developed from mevastatin, a polyketide produced by the fungus Penicillium citrinum [50] (Figure 4).

Epibatidine is a chlorinated alkaloid that is secreted by the Ecuadorian frog Epipedobates anthonyi and poison dart frogs from the Ameerega genus. It was discovered by John W. Daly in 1974. This alkaloid is widely considered an analgesic agent and under active investigation as promising drug leads. A number of natural product drugs are used to treat hypertension and congestive heart failure. These include the angiotensin-converting enzyme inhibitor captopril which is based on the peptidic bradykinin potentiating factor (teprotide) and isolated from venom of the Brazilian arrowhead viper (*Bothrops jararaca*).

Conclusion

Compounds isolated from nature have long been known to possess biological activities and pharmaceutical potential far greater than anything made by man. Natural products represent an untapped source of novel and structurally diverse chemo types with potential to act via novel mechanisms of action against various causative agents of different diseases. This article highlighted the immense therapeutic potential of some natural products isolated from plant, microbial and marine sources. Natural products have played a central role in advancing synthetic and biosynthetic chemistry, medicine, and our understanding of nature. However, natural products are notoriously cumbersome to isolate and very challenging to synthesize. In recent years, several technological and scientific developments including improved isolation, synthesis, analytical tools, and microbial culturing advances are addressing such challenges and opening up new opportunities in the drug discovery process.

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